WIVOZA · LESSON DEBRIEF

Cellular respiration: where a cell's energy comes from • 3

Biology · Oct 14, 2026 at 10:05 AM · 9th · 22:38

NEXT LESSON, TRY THIS

Your "what if the cell had no oxygen?" question got students reasoning.

After a student answers, ask one more "why?" before you move on.

W The lesson covers cellular respiration — how cells turn glucose and oxygen into usable energy — moving from a modelled walk through the equation to small groups predicting what happens when oxygen runs short.

How the lesson moved

OPENED WITH	BUILT	APPLIED TO
Recapping photosynthesis 0:00–7:09	Walking through the respiration equation 7:09–10:48	Groups predicting a cell short of oxygen 10:48–end

YOU SPOKE	STUDENTS SPOKE	SOUND, NO WORDS	QUIET
34.3 %	13.5 %	41.3 %	10.9 %

PACE	QUESTIONS	TAWT	CHECKS
214 wpm	7	7.4 s from 4 questions	1

How to read these numbers ▾

You spoke, Students spoke · the share of the recording each voice took. The microphone hears you better than students.

Quiet · no sound at all: quiet work, pauses, a settled room.

Questions · the questions you asked, as the transcript caught them.

Checks · the times you checked for understanding, as the transcript caught them. It counts the checks, not whether students understood.

Sound, no words · sound the microphone caught but couldn't turn into words, usually voices too far away or talking over each other.

Pace · your words a minute, short pauses included. Everyday conversation usually runs about 120–160.

TAWT (Teacher-Anchored Wait Time) · after a question that asked students to explain or reason, how long until you spoke again, averaged. Three seconds or more gives most students time to think.

KEEP DOING

4 of your questions asked students to explain why or how.

"Why do you think the cell needs oxygen at the very end of the chain?" 18:47

TRY NEXT TIME

The rest of what this report suggests, alongside your one focus.

MOMENT TO REVISIT

Ask it, give pairs thirty seconds to write a one-line guess, and take two answers before you explain.

TALK & PARTICIPATION

Before explaining the group task, ask pairs to predict which stage stops first without oxygen, then build your explanation around their answers.

QUESTIONING & CHECKING

After the yeast question, add one follow-up "why does that make the dough rise?" before moving on.

WHAT WORKED · CLARITY & CONTENT

Building the equation up one molecule at a time — glucose in, oxygen in, then asking where the carbon goes — makes reactants and products something students track rather than copy down.

1

Talk & Participation

Who was heard, and for how long.

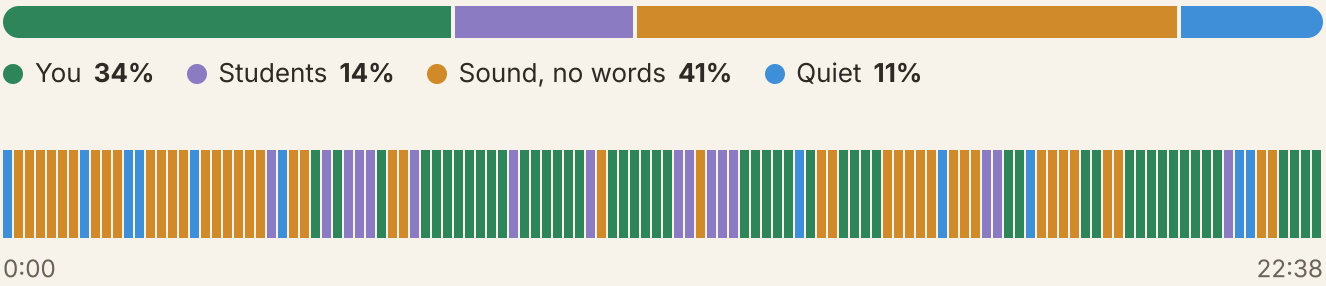
W

Teacher talk ran well ahead of student talk in what the microphone caught, with several extended stretches of you explaining the equation, the three stages, and the group task — explanatory talk rather than rapid-fire pacing.

Students got the floor mainly through a couple of read-aloud moments and brief responses to direct questions; what else was said in group work during the prediction task is likely underrepresented here, since small-group talk is hardest for a single microphone to pick up. This doesn't tell us whether the room was quiet or simply out of the mic's reach — only you know which.

You	Students	Your pace
34%	**14%**	**214**
of the recording	that came through as words	words a minute · conversation usually runs 120–160

Your recording, minute by minute 22:38

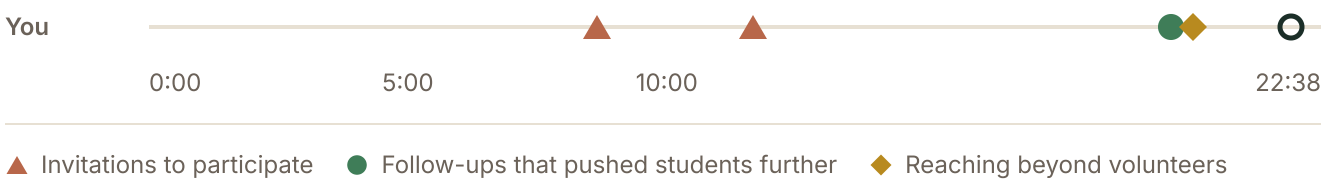


"Sound, no words" is sound the microphone caught but couldn't turn into words. "Students" is speech the transcript didn't attribute to you — a student close to you can be heard as you, and you as them.

HOW YOU BROUGHT STUDENTS IN

4 moves to bring students in — 1 of 4 came in the first half.

When it happened 22:38



○ Check for understanding

▲ 2

Invited the room in

Asking students to share, read, or answer.

● 1

Asked students to say more

Follow-ups that pushed for reasons or detail.

○○○●○●○●○

Followed up on **3 of 9** student answers the recording could hear

◆ 1

Reached beyond volunteers

Brought in students who hadn't raised a hand.

Every moment, in order

8:38	▲	HANDS UP "Can somebody read the equation on the board out loud for us?"
11:40	▲	TURN AND TALK "Turn to your neighbor and tell them which stage you think stops first."
19:44	●	ASKING FOR REASONING "With that in mind, what happens inside a muscle cell during a sprint?"
20:10	◆	BEYOND VOLUNTEERS "Let's hear from a group we haven't heard from yet."

KEEP

You asked a student for their reasoning, not just the answer.

"With that in mind, what happens inside a muscle cell during a sprint?" 19:44

TRY NEXT TIME

Before explaining the group task, ask pairs to predict which stage stops first without oxygen, then build your explanation around their answers.

2 Questioning & Checking

What you asked, how you checked, and how you responded.

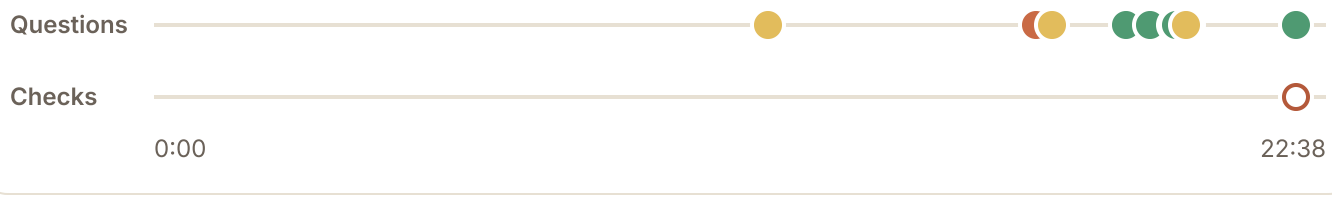
W Of the 33 questions asked, several were invitations to read or volunteer rather than probes into content, and at least one placed the thinking back on you rather than leaving it with students — after asking “So which two go in? Which two?” you supplied the answer yourself rather than waiting it out.

With wait time averaging under three seconds, there’s limited room for a student to gather a fuller response before the question moves on or gets answered by you. It’s not clear from the recording how many of the 33 pushed into reasoning versus simply directing attention or inviting a reader.

Your questions and checks

6 questions · 1 check

4 of 6 were higher-order questions (explain or reasoning). Once you answered your own question. You checked for understanding 1 time.



Recall · 2 of 6

One right answer, retrieved from memory.

- “[name], what do you think the word aerobic means?” 19:41
- “Yeah. Without oxygen the cell switches to what?” 19:58

Explain · 4 of 6

Students put an idea in their own words — why or how.

- “Why do you think the cell needs oxygen at the very end of the chain?” 18:47
- “So which of the three stages would stop first if the oxygen ran out?” 19:14

“With that in mind, what do you think happens inside a muscle cell during a sprint, or what’s the most likely thing it does?” 19:44

“So who can explain why yeast makes bread dough rise?” 22:02

Answered yourself · 1

You asked, and gave the answer before a student could.

“So which two go in? Which two?” 17:00

Follow-up · 3

Right after a student answered, you pushed on what they said.

“Why do you think that is?” 11:52

“So what does aerobic mean” 17:20

“Can you say more about where the carbon goes?” 19:56

Checks for understanding · 1

Moments you asked students to show or say what they understood. No response to the answers was heard.

Check for understanding 22:02

“So who can explain why yeast makes bread dough rise?”

Teacher-Anchored Wait Time (TAWT)

7.4 s average of 4

After each question that asked students to think — to explain or to reason — the time from the end of your question until you spoke again, with any student answer included. It shows how long you left the thinking with the class before stepping back in. When no one answered, three seconds or more of silence gives most students time to think.

Explain · 18:47

“Why do you think the cell needs oxygen at the very end of the chain?”

18.5 s
a student answered

Explain · 19:14

“So which of the three stages would stop first if the oxygen ran out?”

3.1 s
a student answered

Explain · 19:44

“With that in mind, what do you think happens inside a muscle cell during a sprint, or what’s the most likely thing it does?”

3.6 s
a student answered

Explain · 22:02

“So who can explain why yeast makes bread dough rise?”

4.4 s
a student answered

Also heard, not counted as questions: 1 “right?” / “okay?” tag

“right?” 14:23

KEEP

4 of your questions asked students to explain why or how.

“Why do you think the cell needs oxygen at the very end of the chain?” 18:47

TRY NEXT TIME

After the yeast question, add one follow-up "why does that make the dough rise?" before moving on.

Goal: 5 of 9 answers followed up, up from 3.

3

Clarity & Content

What the lesson said it was about, in its own words.

How the lesson moved

OPENED WITH	BUILT	APPLIED TO
Recapping photosynthesis 0:00–7:09	Walking through the respiration equation 7:09–10:48	Groups predicting a cell short of oxygen 10:48–end

From your lesson

2 moments

What you did to make the content clear, in your own words.

Examples & explanations · 1

How you introduced the content and made it approachable — examples, analogies, meanings, thinking aloud.

Gave an example — 5:19

“So the glucose gets split in half first, before any oxygen is even involved, and that split is glycolysis.”

Real-world connections · 1

Links you made to other texts, lessons, or students’ lives.

Connection — 7:00

“Think about how your legs burn at the end of a sprint in gym.”

Academic vocabulary

OPTIONAL

3 of 5 key terms were heard from students.

● You ● Students Each dot = one time said

STUDENTS USED THESE · 3

glucose	●●●●●●●●●	●●
mitochondria	●●●●●●●	●●●

ATP	●●	●

ONLY YOU SAID THESE · 2

electron transport chain	●●●●●	not heard
fermentation	●●●	not heard

KEEP

Building the equation up one molecule at a time — glucose in, oxygen in, then asking where the carbon goes — makes reactants and products something students track rather than copy down.

“it ends up as carbon dioxide. Very good. So the glucose gets split in half first, before any oxygen is even involved,…” 5:19

TRY NEXT TIME

The sprint example is a strong bridge to fermentation — next time you could have groups predict what a muscle cell does when oxygen runs short before you name lactic acid, so the idea comes from their reasoning rather than your explanation.

4

Climate & Routines

Directions, praise and redirections, as they sounded.

W

Transitions were handled directly: when conversation got out of hand during group work, you named it plainly (“Voices off while I explain”) and moved the class along rather than letting it stall the lesson. Directions were layered clearly, naming the task, steps, and timeframe together for the prediction task.

You called on students by name repeatedly and opened the floor for volunteers to read aloud, which invites participation beyond just the loudest voices. Quieter encouragement or a settling look wouldn’t show up here, so this is only what carried to the mic.

From your lesson

11 moments

How your words set up the work, kept it on track, and encouraged students.

Directions & routines · 1

Your one set of directions named what to do, the steps, how long.

Directions for starting work, changing activities, materials, getting help and finishing — and which of the four parts each named.

Named what to do, the steps, how long — 7:54

“So here's what we're going to do. I'll hand out the cell cards in a second. Your group reads the three stages, then decides which one stops first when the oxygen runs out, and why. You've got ten minutes, and then one person from each group reports back.”

Repeated redirections · 7

Your words correcting behavior, each filed under what it was about. One correction tells little; the same one coming back is worth a second look.

Repeated reminders about talking

“You two, if I have to stop again for the side conversation, you’ll be moving seats.” 13:14

Across the lesson, from 9:47 to 20:16, you returned several times to asking students to stop talking.

More evidence ▾

“I don’t want to start moving seats.” 13:55

“Voices down, everybody.” 20:16

No speech was captured from 0:00 to 4:29.

Behavior corrections in this lesson

Talking 9:47
Too short or unclear to quote on its own.

Talking 13:14
“You two, if I have to stop again for the side conversation, you’ll be moving seats.”

Getting back to work 13:45
“This part is on your own, please.”

Talking 13:55
“I don’t want to start moving seats.”

Staying in seats 18:25
“Folks, why are we up and walking around?”

Talking 20:16
“Voices down, everybody.”

Attention 22:16
“Eyes up front, back table.”

Encouragement & respect · 3

1 of the 2 times you responded to a student named what was right or wrong.

What your words recognized, invitations to ask for help, and how you responded to mistakes.

Named what was right 19:21
“Yeah. Carbon dioxide. So the carbon leaves as a gas you breathe out.”

Responded to a mistake 17:23
“No. Light has nothing to do with it. That’s photosynthesis, not this.”

General praise 19:44
“Lactic acid. Very good.”

Your directions, part by part

OPTIONAL

0 of 1 named all four parts.

	What to do	Steps	How long	Hand in
7:54	■	■	■	

A filled square is a part the direction named. Hover a row for its words.

KEEP

You named what was right, not just “good”.

“Yeah. Carbon dioxide. So the carbon leaves as a gas you breathe out.” 19:21

This report reflects what could be heard in your recording — talk patterns, questioning, and classroom routines. It doesn't capture lesson planning, materials, physical space, visual engagement, or anything outside class time. Automated counts above are suggestions to confirm or edit, not final judgments. Audio is never saved: it's sent once for transcription and discarded, and only the transcript and these insights are kept.